



**AI Engineering Techniques
& Architectures Project:**
Meridian Strategy Group
People & Travel Operations Assistant

Solution Design & Evaluation

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AI Disclosure

- Generative AI tools were used during this project as part of the engineering workflow. OpenAI ChatGPT assisted with planning, drafting and review during the project's development.
- The author remains responsible for the project concept, requirements, architecture, implementation decisions, validation, evaluation and submitted content.
- AI-generated outputs informed the design and development process but did not determine the final project decisions.
- Each recommendation was assessed and revised using the project requirements, implementation evidence and observed system behaviour.
- ChatGPT assisted with defining and refining the project concept, use case, business requirements and software requirements. It also helped compare architecture and design options, identify edge cases and evaluation considerations, improve requirements traceability, explain technical concepts such as retrieval-augmented generation (RAG), agent orchestration, Model Context Protocol (MCP), deployment, CI/CD and evaluation, and review project artefacts and documentation.
- AI coding tools assisted with code generation, debugging, testing and repository-level verification. All generated code and technical recommendations were reviewed, tested and adjusted before use.
- The accompanying AI Tooling documentation identifies the tools used, their roles and the relevant AI interactions.

Executive Summary

Company Background and Business Context

Meridian Strategy Group (**MSG**) is a global management consulting organisation comprising both client-facing consulting professionals and enablement functions such as HR, Finance, IT, Legal/Risk and Operations. Client-facing employees regularly work across client engagements and locations, creating a highly mobile workforce whose time is directly connected to client delivery and revenue generation.

While on assignment, employees still need to resolve routine administrative matters such as leave, travel changes, expenses, benefits and approvals. These requests can become surprisingly complex because the correct answer may depend simultaneously on the employee's role, active assignment, travel authorisation, operational records and the applicable company policy or procedure.

For a consulting organisation, the issue is therefore not simply access to HR policies. Employees need to resolve administrative questions quickly, accurately and in the context of their current work, without unnecessarily interrupting client delivery or relying on multiple support teams and systems.

Agent Overview

Meridian Compass is a fictional internal employee assistant developed for Meridian Strategy Group (MSG). It supports employees whose administrative questions may depend on several factors at once, including their identity, current assignment or travel context, approved company policy, relevant procedures and the appropriate approval relationship.

The challenge is therefore not simply finding a policy document. Employees must also understand how the relevant rule applies to their current circumstances and determine the correct next step. Meridian Compass addresses this need by combining approved internal guidance with authorised operational information to provide a clear, employee-specific response supported by cited evidence.

Design Objective

Meridian Compass combines authenticated identity, current operational context and approved internal knowledge so that answers are employee-specific, cited and auditable. The language model synthesises supported evidence; it is not treated as a source of company policy.

Logical Architecture

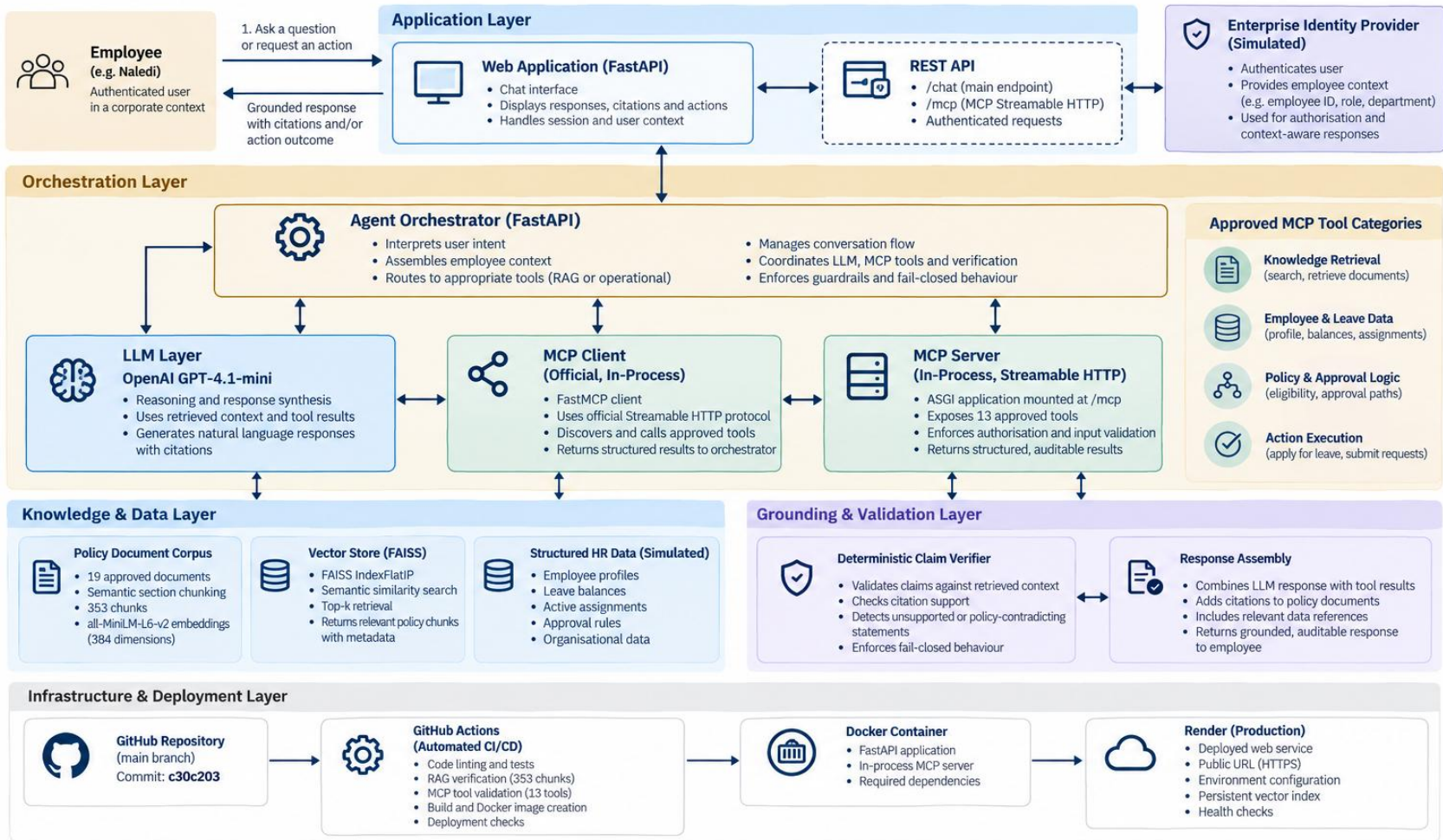
Employee -> Web UI / FastAPI -> Simulated Enterprise Identity -> Agent Orchestrator -> MCP Client -> Streamable HTTP -> MCP Server -> RAG knowledge and structured/mock tools -> Deterministic Claim Verifier -> Answer + citations + snippets + trace. The LLM provider is called through an abstraction and remains external to the application container.

Figure 1. Master Architecture

Meridian Compass

Overall Solution Architecture

Meridian Compass is a secure, enterprise-grade AI assistant that provides grounded, policy-aligned guidance and executes authorised HR-related actions using RAG and MCP.



Architecture Responsibilities

Layer	Responsibility
Web/API	Authentication entry, chat experience, POST /chat and GET /health
Identity	Server-authoritative simulated employee identity and bounded session context
Agent orchestration	Intent routing, approved tool selection, evidence assembly, synthesis
MCP boundary	Protocol-mediated access to knowledge, structured reads and controlled actions
RAG	Retrieve approved policy/procedure evidence with metadata
Structured operations	Return authorised synthetic employee, PTO, assignment, travel, expense and benefit facts
LLM provider	Interpret/synthesise bounded evidence
Claim verifier	Deterministically reject unsupported material claims/citations

Controlled Orchestration

- Authenticate.
- Classify and route the request.
- Invoke planner-selected approved MCP tools.
- Gather structured facts.
- Retrieve policy/procedure evidence.
- Resolve approval relationships where needed.
- Synthesize using bounded evidence.
- Verify material claims.
- Return grounded answer or fail closed/clarify/escalate only when supported.

RAG Design

Element	Design
Corpus	12 policies + 7 procedures = 19 approved documents
Chunking	Semantic section chunks with metadata
Embedding model	all-MiniLM-L6-v2
Vector representation	384-dimensional normalized embeddings
Vector store	FAISS IndexFlatIP
Index size	353 chunks
Baseline retrieval	k=5; comparison at k=3/5/8

Runtime assurance	Artifact presence, chunk count/fingerprint, corpus fingerprint, model name/init checks
Grounding	Material policy/procedure claims require extractive evidence

MCP Design

Transport is official Streamable HTTP. The deployed single-service architecture physically co-locates web/API, orchestrator, MCP client/server, RAG/index and synthetic data while preserving the logical MCP protocol boundary. The orchestrator does not directly read MCP-owned operational JSON.

Category	Tools
Knowledge	search_knowledge_documents
Structured reads	lookup_employee_profile; check_pto_balance; lookup_benefits_status; lookup_active_assignment; lookup_travel_authorization; get_mock_travel_booking; get_per_diem_rate; get_mock_expense_claim; resolve_approval_role
Controlled actions	draft_hr_email; create_mock_hr_ticket; create_mock_travel_request

Demonstration Workflow: Planned PTO

For a request to take PTO on 15-16 October 2026, the production system combined authenticated identity, employee profile, 15 available PTO days, an active Nairobi assignment, leave policy evidence and approval resolution. At runtime on 4 October, the request had 11 calendar days' notice versus the standard 14-day planned-PTO requirement. Amara Okafor was resolved as the recorded Engagement Manager. The response correctly distinguished permission to submit from actual approval and cited MSG-POL-001.

Demonstration Workflow: Personal Travel Extension

The controlled fixture compares a business-equivalent fare of ZAR 8,400 with a proposed extended return fare of ZAR 9,650, leaving ZAR 1,250 incremental personal cost. The workflow combines travel authorisation, booking, assignment/PTO context, per-diem/travel policy and approval routing. Policy constrains hotel/per-diem treatment and requires approval before itinerary modification.

Evaluation

Measure	Result
Controlled cases	27/30 passed (90%)
Remaining failures	1 literal eight vs 8 expectation; 2 expense prompts omitted required expense_id while expecting lookup
Retrieval comparison	k=3, k=5 and k=8 all 14/30 at earlier stage; dominant issue was routing, not retrieval depth
Ablation	PASS: removed policy caused insufficient evidence rather than fabrication
Latency	Mean 168.318 ms; p50 126.106 ms; p95 355.581 ms (local deterministic, not live LLM latency)
Automated tests	138 collected; final GitHub Actions passed on Python 3.12

Deployment and Operational Learning

The Docker-based Render deployment exposes a healthy /health response and completed live PTO production acceptance. A 512 MB instance could start but exceeded memory during a full agent workflow. The same application build passed production acceptance after right-sizing to 2 GB RAM. This is recorded as an infrastructure resource constraint, not hidden as an application-code fix.

Key Engineering Conclusion

Meridian Compass treats the LLM as one component of a controlled system. Authenticated identity, authorised operational context, approved RAG evidence, MCP-mediated tools and deterministic verification combine to turn a generic employee question into a grounded employee-specific next action.